



BHABHA UNIVERSITY

DEPARTMENT OF ELECTRONICS AND COMMUNICATION
M.TECH III SEM, VLSI DESIGN

Scheme Of Examination

S.No.	Subject Code	Subject Name	Periods per week			CREDITS	Maximum Marks (Practical Slot)		Total Marks		End Sem. Practical /Viva	Practical Record/ Assignment/Quiz /Presentation	TOTAL MARKS
			L	T	P		End Sem. Exam.	Tests (Two)	Assignments/ Quiz				
1.	EC-305	Optoelectronics Integrated Circuits	3	1	-	4	70	20	10	-	-	100	
2.	EC-306	Communication RF IC Design	3	1	-	4	70	20	10	-	-	100	
3.	EC-307	Seminar	-	-	4	4	-	-	-	-	100	100	
4.	EC-308	Dissertation part 1	-	-	8	8	-	-	-	120	80	200	
		Total	6	2	12	20	140	40	20	120	180	500	

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SYLLABUS

EC – 305 Opto-Electronics Integrated Circuits

UNIT I

Theory of Optical Wave guides: Wave guide theory : one dimensional planar wave guides, two dimensional wave guides, transcendental equations, wave guide modes, mode cutoff conditions.

UNIT II

Optical Wave guide Fabrication and Characterization: Waveguide fabrication: deposited films; vacuum-deposition and solution-deposition, diffused waveguides, ion-exchange and ion-implanted waveguides, epitaxial growth of III-V compound semiconductor materials, shaping of waveguides by wet and dry etching techniques. Waveguide characterization: surface scattering and absorption losses, radiation and bending losses, measurement of waveguide loss, waveguide profiling.

UNIT III

Fundamentals of Optical Coupling: Transverse couplers. Prism couplers. Grating couplers. Fiber to waveguide couplers. Coupling between optical waveguides. Directional couplers. Applications of directional couplers.

UNIT IV

Guided Wave Modulators and Switches: Physical effects used in light modulators : electro-optic, acousto-optic and magneto-optic effects. Waveguide modulators and switches.

UNIT V

Semiconductor Lasers and Detectors: Laser diodes. Distributed feedback lasers. Integrated optical detectors.

UNIT VI

Recent Progress in Integrated Optics: State-of-the-art technology in guided wave devices and applications, e.g. photonic switching, tunable laser diodes, optical integrated circuits.

TEXT/ REFERENCE

1. T Tamir, *Guided Wave Optoelectronics*, Springer-Verlag, 1990
- R Sysm & J Cozens, *Optical Guided Waves and Devices*, McGraw

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EC -306 Communication RF IC Design

UNIT I

Basic concepts in RF Design: Analysis & Measurement Techniques. S-Parameter Models, Smith Chart Calculations.

UNIT II

Trans-receiver Architecture for Wireless Communication Standards. Non-Linearity, Harmonics, Gain Compression, Desensitization, Cross Modulation, IMD & Inter-symbol Interface.

UNIT III

RF IC Design concepts & Device Technologies: Low Noise Amplifiers, Mixers, Frequency Sources, Oscillators & Synthesizers, Power Amplifiers. Noises & Distortions in LNA, PA & Mixer Circuits.

UNIT IV

PLL: Theory, Circuits, Distortion & Noises. Microwave Circuit Components & Design Concepts: Single Chip Radio Concepts, Design Issues Surrounding Systems as DECT, GSM, Blue Tooth etc. Case Studies.

TEXT / REFERENCE

1. Behzad Razavi, *RF Microelectronics*, PHI 1998.
2. R. Ludwig & P. Bretchko, *RF Circuit Design*, PHI 2000.
3. L.E. Larson, *RF & Microwave Circuit Design for Wireless Communication*, Artech House Publishers, 1997.
4. Thomas H. Lee, *The Design of CMOS Radio Frequency Integrated Circuits*, Cambridge University Press, 1998.
5. George Vendelin, *Design of Amplifiers & Oscillators by S-Parameter Method*, J. Wiley & Sons, 1982.

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